COMPANY B

WEB DEVELOPMENT COMPANY

Smart Bus Ticketing and Reservation System

Software Requirements Specification

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) and Design Document outlines the functional, non-functional and architectural specifications for the Smart Bus Ticketing and Reservation System (SBTRS). The goal of SBTRS is to digitize reservations for long-distance buses using a prepaid wallet mechanism ensuring secure and reliable booking.

1.2 Scope

The system enables:

- Booking tickets for predefined routes based on distance and sectioning.
- Managing a digital prepaid wallet for all transactions.
- Viewing analytics dashboards for Passengers, Drivers, Bus Owners, and Admins.
- Implementing automated refund handling and ticket disbursement.

Out of Scope:

- Real-time GPS tracking of buses.
- Support for undefined custom routes or ad-hoc short-distance passengers.
- Mobile native application (web-only system).

2 Stakeholders & Roles

2.1 End Users

- **Passengers**: Book tickets, top-up wallet, submit refund requests.
- **Drivers**: View passenger manifest and verify bookings.
- **Bus Owners**: View earnings, utilization, performance analytics.
- **Administrators**: Manage routes, users, disputes and refunds.

2.2 During Project Development

- **Product Owner/Client**: Defines business logic and validates deliverables.
- **Development Team**: Responsible for design, implementation, testing, deployment.

3 Requirements Specification

3.1 Functional Requirements

Ticket Provisioning

- System calculates ticket prices based on distance/sections.
- Issues unique digital tickets tied to user ID.

Prepaid Wallet

- Supports wallet top-up via payment gateway.
- Handles fare deductions and automated refunds.

Unique ID Booking

- Tickets must be linked to a unique customer ID.
- System must prevent duplicate bookings and ticket scalping.

Role-Based Dashboards

- Passenger Portal
- Driver Portal
- Owner Dashboard
- Administrator Control Panel

3.2 Non-Functional Requirements

Refund Policy

- Robust automated refund decision flow.
- Manual override by administrators.

Scalability

- Must support regional deployment initially, with island-wide scalability.

Availability

- 99.9% uptime.
- 24/7 access for bookings and modifications.

4 Architectural Design

4.1 High-Level Architecture

The system uses a modular architecture comprising:

- User Authentication & ID Verification Module
- Wallet & Payment Module
- Ticketing & Route Engine
- Refund & Dispute Manager
- Analytics Engine for Owner Dashboard

4.2 Key Module Descriptions

Wallet & Payment Module

- Handles top-ups, fare deductions, and refund settlements.

Ticketing Engine

- Manages seat selection, pricing, sectioning.

Dispute & Refund Manager

- Accepts passenger claims.
- Routes decisions to admin review when required.

5 UML Class Diagram

Below is the UML Class Diagram representing core entities of SBTRS.

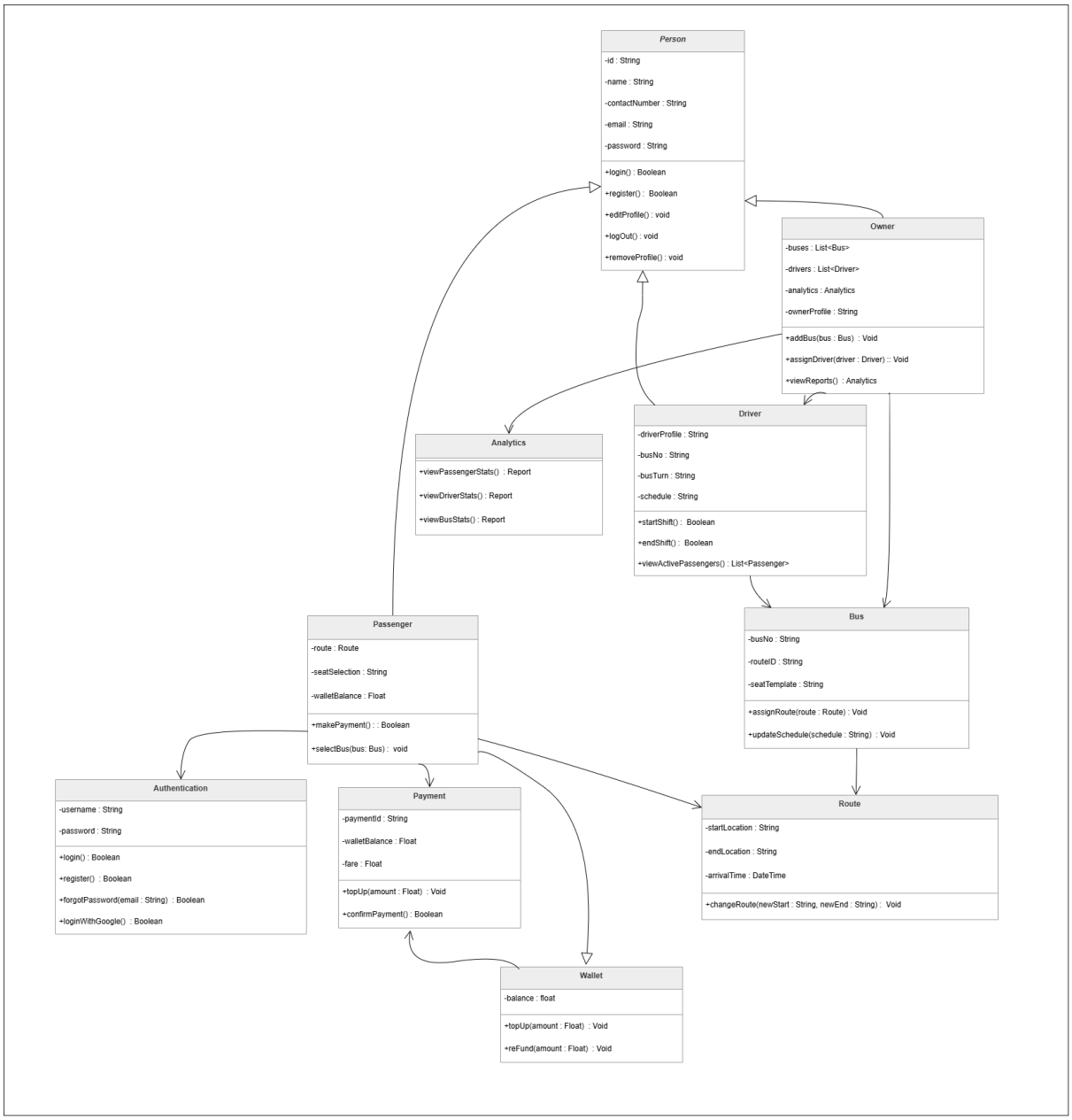

Figure 1: UML Class Diagram for Smart Bus Ticketing and Reservation System

6 Project Management & Modularity

6.1 Phase Breakdown

Phase 1 (Weeks 1-2)

- Database schema
- Authentication & ID verification

Phase 2 (Weeks 3-5)

- Prepaid wallet implementation
- Route and distance pricing engine

Phase 3 (Weeks 6-7)

- Ticketing engine
- Refund workflow
- All dashboards

Phase 4 (Week 8)

- User Acceptance Testing
- Load testing
- Deployment

7 Deliverables & Success Metrics

7.1 Deliverables

- Fully deployed web application
- Source code repository with documentation
- User Manuals for all roles

7.2 Success Metrics

- Supports 500+ concurrent wallet transactions
- 0% duplicate bookings due to ID verification
- 100% functional requirements delivered

8 Sign-Off & Evaluation Confidence

By adhering strictly to scope constraints:

- No GPS tracking
- Web-app-only design
- Centralized prepaid wallet

The architecture mitigates major risks and ensures scalability and reliability (24/7 availability and island-wide scalability).

9 Approval & Sign-Off

Signature: **Date:**

27/03/2026

Company A Head
Smart Bus Ticketing and Reservation System (SBTRS)

By signing this document, the Head of Company A formally approves the Software Requirements Specification (SRS) as the binding contractual reference for the Smart Bus Ticketing and Reservation System project. This approval confirms agreement to the scope, requirements, and deliverables outlined herein, and authorizes the development team to proceed with design, implementation, and deployment in accordance with the terms of the web contract.

10 References & Resources

Repository

The source code and documentation are maintained in the following GitHub repository: Smart Bus Ticketing and Reservation GitHub Repo

Design Prototype

The UI/UX design prototype is available on Figma: Smart Bus Ticketing Figma Design